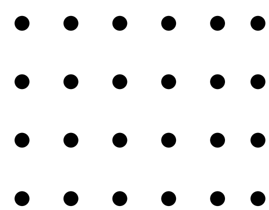
Unknown Factors and Related Division Facts

Grade 2–3 — Foundations of Division

Focus: Arrays, unknown-factor equations, equal-group stories, and related division facts.

Directions: Every problem hides one factor. Write the multiplication equation with a box or a question mark for the missing factor, think "what times this gives the total?", then write the factor you found on the answer line. Draw or count if it helps.

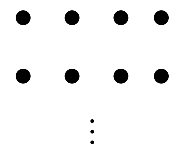
1. Maya lines up her stickers in **4 equal rows**. She has **24** stickers in all.



Fill in the missing factor: $4 \times \_\_\_\_ = 24$.
How many stickers are in each row?

Answer: __________________

2. A gardener plants **28** seeds. Each row has **4** seeds, and every row is full.



(Only the first two rows are drawn.)
Fill in the missing factor: ____ $\times 4 = 28$.
How many rows of seeds are there?

Answer: ________________

3. Find the unknown factor. Say the times table you are using out loud.

$$7 \times \_\_\_\_ = 56$$

What number belongs in the blank?

Answer: ________________

4. Nina has **45** stickers. She shares them equally among **5** friends, so every friend gets the same number of stickers.

Write the equation $5 \times \_\_\_\_ = 45$, then find the unknown factor.

How many stickers does each friend get?

Answer: ________________

5. Sam already knows this multiplication fact: $6 \times 8 = 48$.

Use that fact — do **not** start over — to answer the division question below.

$$48 \div 8 = \_\_\_\_$$

What is the quotient?

Answer: ________________

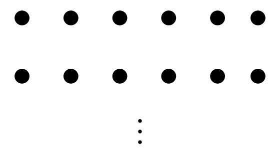
6. Find the unknown factor. The missing number is in front this time.

$$\_\_\_\_ \times 9 = 72$$

What number belongs in the blank?

Answer: ________________

7. A baking tray holds **42** muffin cups in equal rows. Each row has **6** cups. The picture shows only the first two rows.



Write ____ $\times 6 = 42$ and find the unknown factor.
How many rows of muffin cups does the tray have?

Answer: ________________

8. A classroom has **6** tables, and every table has the same number of chairs. Altogether there are **54** chairs.

Write the equation $6 \times$ ____ $= 54$, then solve it.
How many chairs are at each table?

Answer: ________________

9. A mystery number times **7** equals **42**.

(Think first, then compute.) Find the mystery number using the unknown-factor idea, then multiply that same mystery number by **8**.

What is the mystery number times 8?

Answer: ____________

10. A baker has **48** rolls. She puts **6** rolls in each bag, and then packs **2** bags into each box.

Work in two steps: first find how many bags she fills, then find how many boxes she fills.

How many boxes does the baker fill?

Answer: ____________